Delta Physics

Our Dynamic Universe

Special relativity

5 minutes – Individual

Special Relativity

A muon lives for 2.2 μs. How does it cross 10 km of air?



Higher Physics

PART 3

Length Contraction and the Exam

A spacecraft flies past at 0.8c. Is it still the same length?

By the end of this part…



1. I can **state** that a moving object is **shorter** along its motion
2. I can **use** $l' = l\sqrt{(1 - (v/c)^2)}$ to find l', l or v
3. I can **explain** why muons reach the ground, **in both frames**
4. I can **answer** exam questions on **special relativity**

Length Contraction

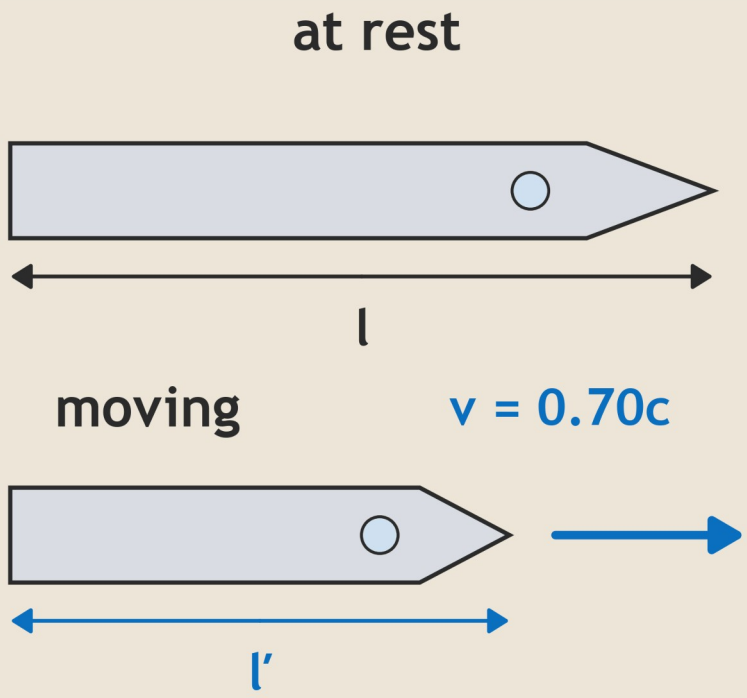
Moving

shorter for a stationary observer

Only

along the motion: height **unchanged**

Try it: length contraction on the website.



l and l’

l

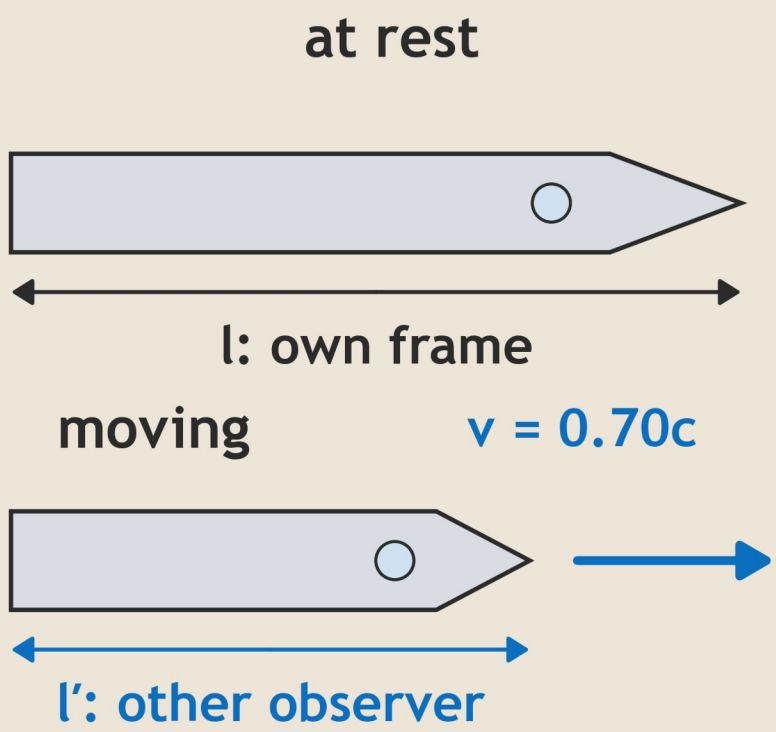
length in its **own frame**: at rest with it

l’

length for the **other observer**

Check

l’ **is shorter** than l



Length Contraction

$$l' = l\sqrt{(1 - (v/c)^2)}$$

l'

observed length (m)

l

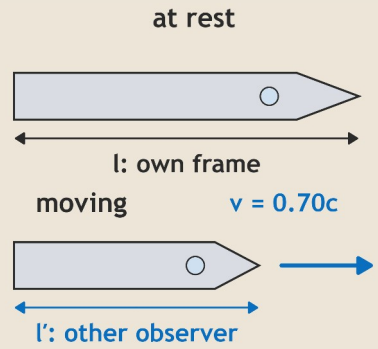
own length (m)

v

speed (m s^{-1})

c

3.00×10^8 m s^{-1}



Worked Example 1

A spacecraft has a length of 80 m in its own frame of reference. It passes the Earth at 0.8c.

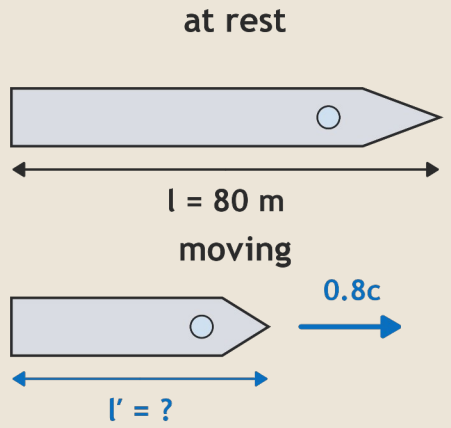
Calculate the length of the spacecraft measured by an observer on Earth. **3**

$$l' = l\sqrt{(1 - (v/c)^2)}$$

$$l' = 80\sqrt{(1 - (0.8c/c)^2)}$$

$$l' = 48 \text{ m}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Worked Example 2

A spacecraft has a length of 100 m in its own frame of reference. An observer on a planet measures its length as 60 m.

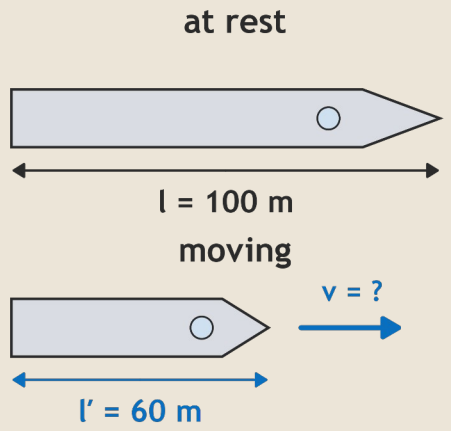
Calculate the speed of the spacecraft relative to the planet. **3**

$$l' = l\sqrt{(1 - (v/c)^2)}$$

$$60 = 100\sqrt{(1 - (v/c)^2)}$$

$$v = 0.8c = 2.4 \times 10^8 \text{ m s}^{-1}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Your Turn 1

A spacecraft has a length of 150 m in its own frame of reference. It travels at 2.0×10^8 m s^{-1} relative to a planet.

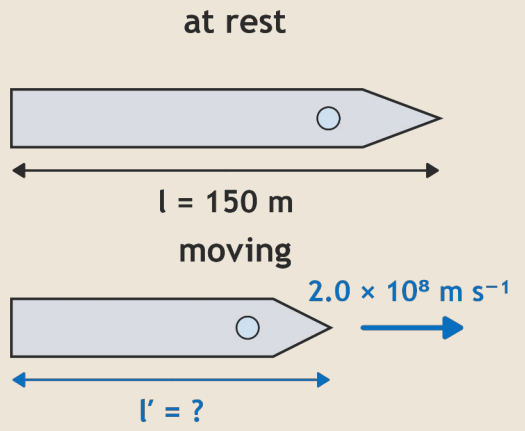
Calculate the length of the spacecraft measured by an observer on the planet. **3**

$$l' = l \sqrt{(1 - (v/c)^2)}$$

$$l' = 150 \sqrt{(1 - (2.0 \times 10^8 / 3.00 \times 10^8)^2)}$$

$$l' = 112 \text{ m}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Your Turn 2: 2023 Paper 2 Q4(c)(ii)

(c) Two physics students on train A are discussing the possibility of travelling at relativistic speeds. The students consider the train travelling at a speed of $0.9c$ relative to a stationary observer.

(ii) Train A has a length of 142 m, as measured in the frame of reference of the students on the train.

Calculate the length of train A when travelling at $0.9c$ as measured by the stationary observer. **3**

Space for working and answer

[Turn over

Scan: page 13.

Plan: 142 m is in the **train's** frame: l

Marking: 2023 Paper 2 Q4(c)(ii)

Question			Expected response		Max mark	Additional guidance
		(ii)	$l' = l\sqrt{1-\left(\frac{v}{c}\right)^2}$	(1)	3	Accept: 62, 61.90, 61.896 Accept: $l' = 142\sqrt{1-(0.9)^2}$
			$l' = 142\sqrt{1-\left(\frac{0.9c}{c}\right)^2}$	(1)		
			$l' = 61.9$ m	(1)		

Past Paper: 2026 Q8

8. An astronaut in a spacecraft measures the length of the spacecraft as 25.0 m.

The spacecraft is travelling at a speed of $0.200c$ as it passes a planet.

An observer on the planet measures the length of the spacecraft as

A 20.0 m

B 22.4 m

C 24.0 m

D 24.5 m

E 25.5 m.

[Turn over

Answer D: $l' = 25.0\sqrt{(1 - 0.200^2)} = 24.5$ m.

Scan: page 7.

Past Paper: 2025 Q8

8. The Large Hadron Collider (LHC) is contained in a circular tunnel.

A technician working at the LHC measures the length of the tunnel to be 26.7 km.

Protons in the tunnel are accelerated to a speed of $0.999c$.

The length of the tunnel in the frame of reference of the protons is

A 0.844 km

B 1.19 km

C 26.7 km

D 597 km

E 844 km.

[Turn over

Answer B: $l' = 26.7\sqrt{(1 - 0.999^2)} = 1.19$ km.

Scan: page 7.

Past Paper: 2019 Q8

8. A spacecraft is travelling at a constant speed relative to a nearby planet.

A technician on the spacecraft measures the length of the spacecraft as 275 m.

An observer on the planet measures the length of the spacecraft as 125 m.

The speed of the spacecraft relative to the observer on the nearby planet is

A $1{\cdot}54 \times 10^{4}\ \text{m s}^{-1}$

B $2{\cdot}22 \times 10^{8}\ \text{m s}^{-1}$

C $2{\cdot}67 \times 10^{8}\ \text{m s}^{-1}$

D $3{\cdot}00 \times 10^{8}\ \text{m s}^{-1}$

E $7{\cdot}14 \times 10^{16}\ \text{m s}^{-1}$.

[Turn over

Answer C: $125 = 275\sqrt{(1 - (v/c)^2)}$, so $v = 2.67 \times 10^8\ \text{m s}^{-1}$.

Scan: page 7.

Past Paper: Specimen Q9

9. A spaceship on a launch pad is measured to have a length L.

This spaceship has a speed of $2{\cdot}5 \times 10^8\ \text{m s}^{-1}$ as it passes a planet.

Which row in the table describes the length of the spaceship as measured by the pilot in the spaceship and an observer on the planet?

	Length measured by pilot in the spaceship	*Length measured by observer on the planet*
A	L	greater than L
B	L	L
C	L	less than L
D	greater than L	L
E	less than L	less than L

[Turn over

Answer C: pilot L (own frame); planet less than L.

Scan: specimen Paper 1, Q9.

Cosmic Rays Make Muons

Cosmic rays

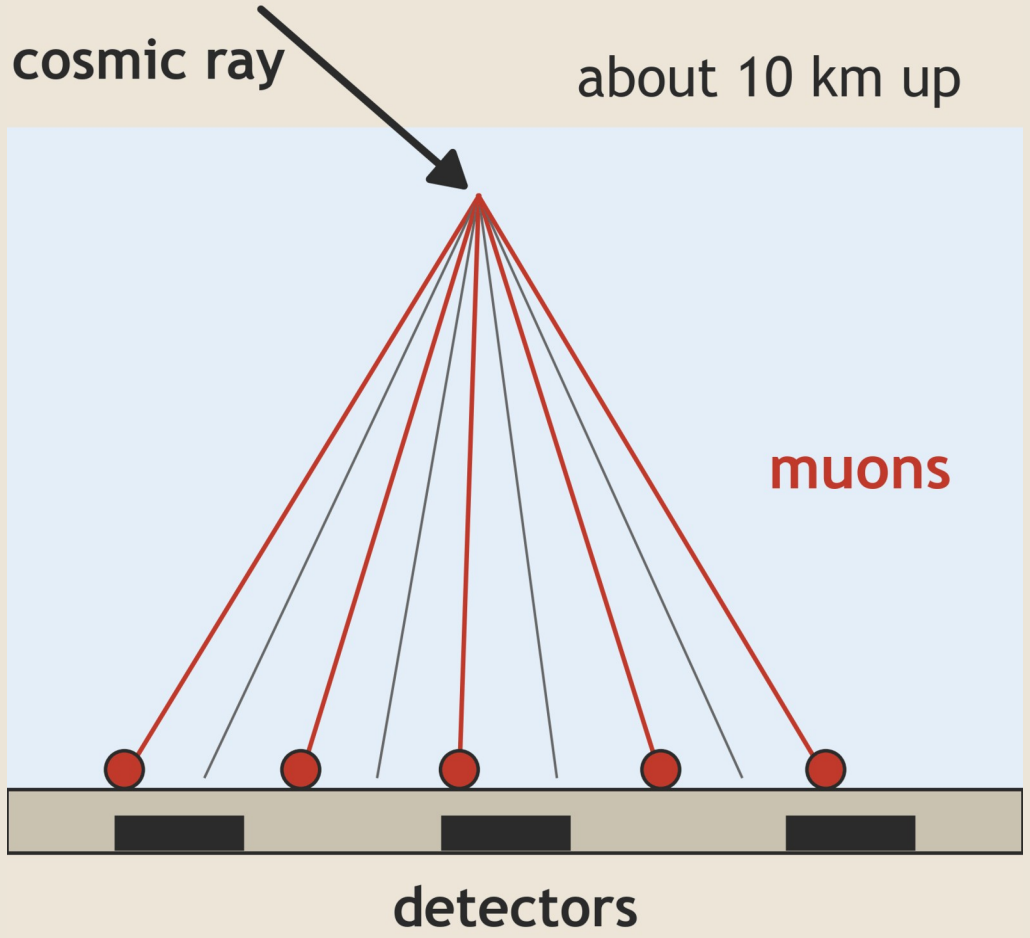
hit the atmosphere about **10 km** up

Muons

travel down at nearly **c**

Lifetime

2.2 μs in their own frame



Counting Muons

Detectors on the ground **count** muons

Far more arrive than a 2.2 μs lifetime allows, **without** relativity



Muons: Two Frames

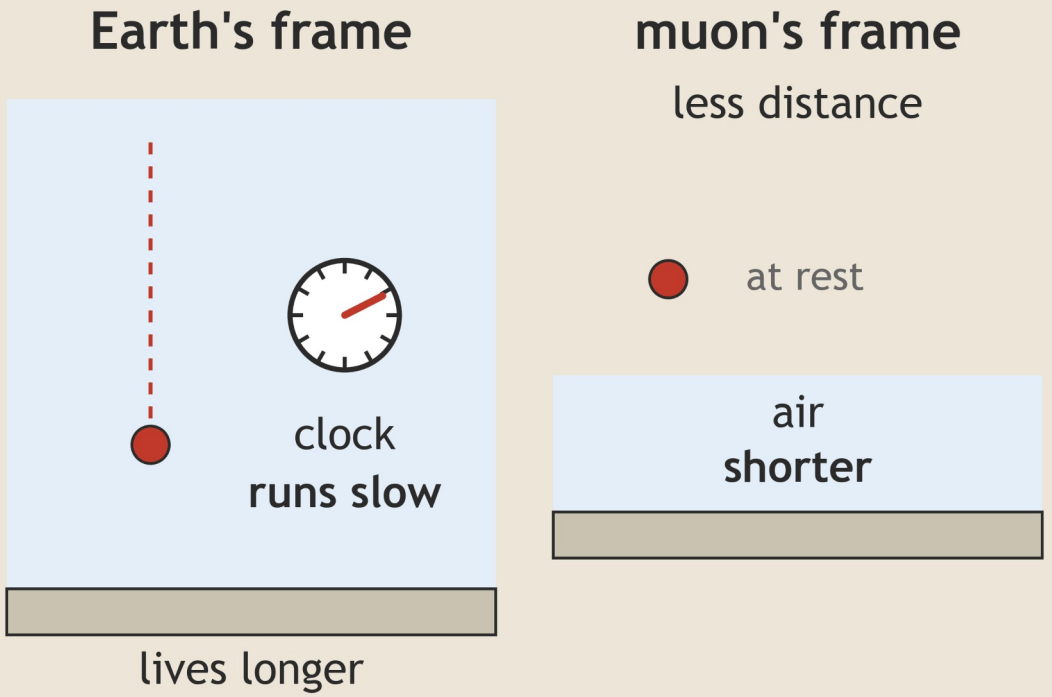
Earth's frame

the muon's **clock runs slow**: it **lives longer**

Muon's frame

the **atmosphere** is **contracted**

Try it: the muons on the website.



Review Questions



1. A train travels at 25 m s^{-1}. A passenger walks forward at 2.0 m s^{-1}. **Determine** the passenger's speed relative to the platform.

27 m s^{-1}

2. A clock on a spacecraft at 0.8c measures 12 s. **Calculate** the time measured on Earth.

20 s

Review Questions



3 A rod is 2.0 m long in its own frame. It moves at 0.6c. **Calculate** its length for a stationary observer.

1.6 m

4 A spacecraft is 50 m long in its own frame and 30 m to an observer. **Calculate** its speed.

$0.8c = 2.4 \times 10^{8}$ m s^{-1}

Explain: Name the Frame

Earth's frame

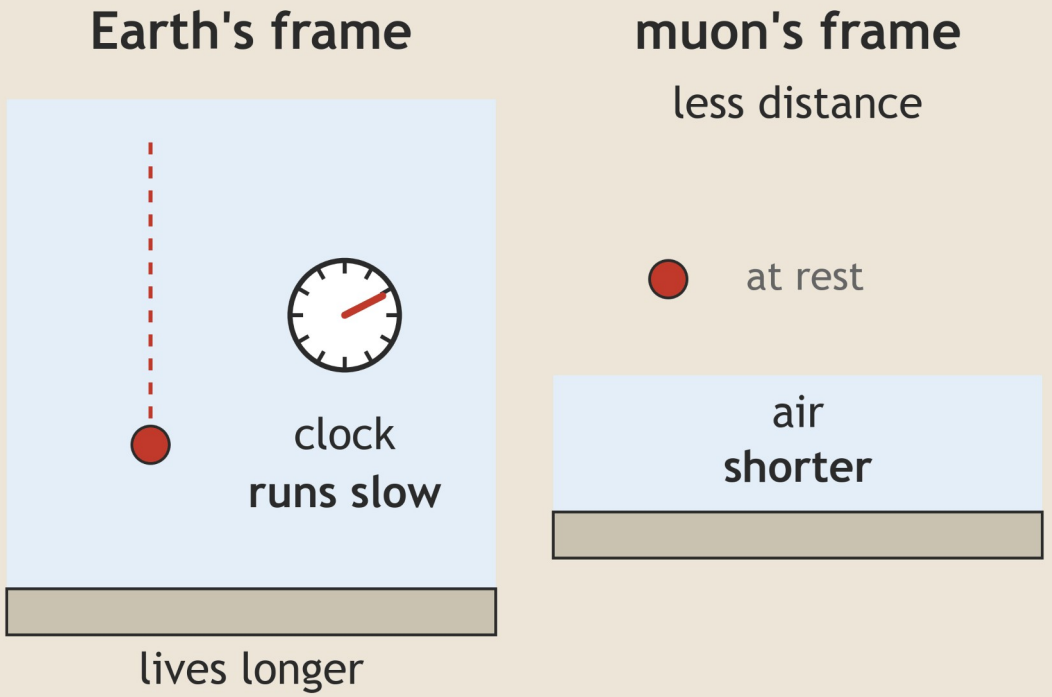
the mean **lifetime** is **greater**

Particle's frame

the **distance** is **shorter**

Either

but say **which frame** (2022 MI Q6(d))



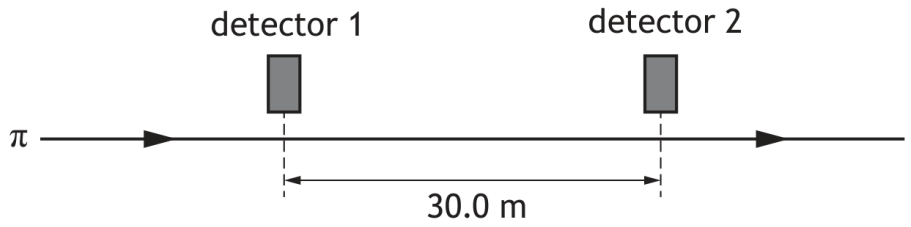
Exam: 2022 Paper 2 Q6(c)(ii)(d)

6. The Standard Model explains how the basic building blocks of matter interact, governed by four fundamental forces.

(c) Another particle, known as a pion (π), is a product of kaon decay.

A beam of pions, travelling in a straight line at a speed of $0.95c$, passes between two detectors.

The detectors are 30.0 m apart as measured by a stationary observer.



(ii) Calculate the distance between the two detectors in the frame of reference of the pions. **3**

Space for working and answer

(d) Pions have a mean lifetime of 26 ns in their frame of reference.

Explain why a greater number of pions are detected at the second detector than would be expected if relativistic effects are not taken into account. **1**

Plan: 30.0 m is the **observer's:** the pions see it **shorter**

Scan: p.21 -22.

Marking: 2022 Paper 2 Q6(c)(ii)(d)

Question			Expected response	Max mark	Additional guidance
		(ii)	$l' = l\sqrt{1-\left(\frac{v}{c}\right)^2}$ **(1)** $l' = 30.0\sqrt{1-\left(\frac{0.95c}{c}\right)^2}$ **(1)** $l' = 9.37$ m **(1)**	**3**	Accept: 9.4, 9.367, 9.3675 Accept: $l' = 30.0\sqrt{1-(0.95)^2}$
	(d)		For a stationary observer's frame of reference, the mean lifetime of the pion is greater (than 26 ns) **OR** In a pion's frame of reference, the distance is shorter (than 30.0 m).	**1**	The response must involve a statement referring to, or implying, a frame of reference.

Exam: Specimen Paper 2 Q4

4. Muons are sub-atomic particles produced when cosmic rays enter the atmosphere about 10 km above the surface of the Earth.

Muons have a mean lifetime of $2{\cdot}2 \times 10^{-6}$ s in their frame of reference. Muons are travelling at $0{\cdot}995c$ relative to an observer on Earth.

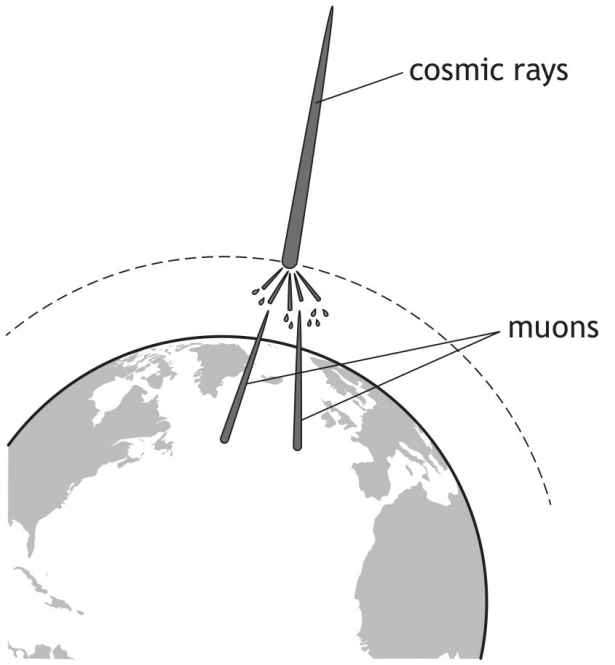
(a) Show that the mean distance travelled by the muons in their frame of reference is 660 m. **2**

Space for working and answer

(b) Calculate the mean lifetime of the muons measured by an observer on Earth. **3**

Space for working and answer

(c) Explain why a greater number of muons are detected on the surface of the Earth than would be expected if relativistic effects were not taken into account. **1**



Scan: specimen Paper 2, Q4.

Marking: Specimen Paper 2 Q4

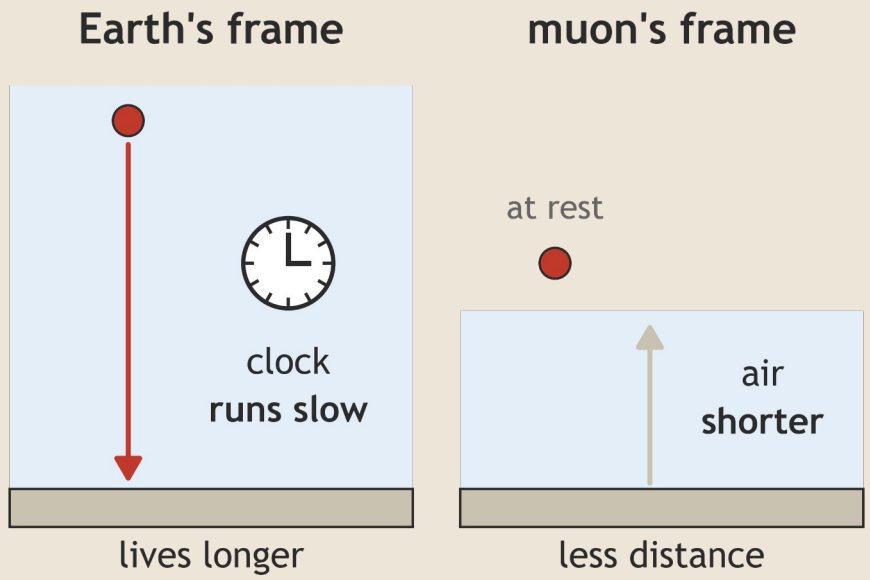
Question			Expected response	Max mark	Additional guidance
4.	(a)		$d = \bar{v}t$ (1) $d = (3{\cdot}00\times10^{8}\times0{\cdot}995)\times2{\cdot}2\times10^{-6}$ (1) $d = 660$ m	2	SHOW question. A maximum of 1 mark is available if the final line is not shown.
	(b)		$t' = \dfrac{t}{\sqrt{1-\left(\dfrac{v}{c}\right)^2}}$ (1) $t' = \dfrac{2{\cdot}2\times10^{-6}}{\sqrt{1-\left(\dfrac{0{\cdot}995}{1}\right)^2}}$ (1) $t' = 2{\cdot}2\times10^{-5}$ s (1)	3	Accept 2, 2·20, 2·203
	(c)		The mean lifetime of the muon is greater for an observer in Earth's frame of reference OR The mean distance travelled by a muon is shorter in the muon's frame of reference	1	

What Candidates Get Wrong (from course reports)

Name the frame: the answer must refer to one (2022 MI Q6(d), est. 31 %)

Why more muons: est. 31 % (2019 SQP Q4(c))

Show that: the final line must be shown (SQP MI Q4(a))



Look back — by the end of this part…



1. I can **state** that a moving object is **shorter** along its motion
2. I can **use** $l' = l\sqrt{(1 - (v/c)^2)}$ to find l', l or v
3. I can **explain** why muons reach the ground, **in both frames**
4. I can **answer** exam questions on **special relativity**

Course specification — 1 of 1

- Knowledge that measurements of space, time and distance for a moving observer are changed relative to those for a stationary observer, giving rise to time dilation and length contraction.
- Use of appropriate relationships to solve problems involving time dilation, length contraction, and speed.

Scan: the course specification, p.8–9.

Credits (1 of 2)

Title: muon detector: Photo: CusterDome, CC0, via Wikimedia Commons

Your Turn 2: 2023 Paper 2 Q4(c)(ii): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4

Marking: 2023 Paper 2 Q4(c)(ii): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4

Past Paper: 2026 Q8: © Qualifications Scotland (SQA). 2026 Higher Physics Paper 1, Q8

Past Paper: 2025 Q8: © Qualifications Scotland (SQA). 2025 Higher Physics Paper 1, Q8

Past Paper: 2019 Q8: © Qualifications Scotland (SQA). 2019 Higher Physics Paper 1, Q8

Past Paper: Specimen Q9: © Qualifications Scotland (SQA). Specimen Higher Physics Paper 1, Q9

Counting Muons: Photo: CusterDome, CC0, via Wikimedia Commons

Credits (2 of 2)

Exam: 2022 Paper 2 Q6(c)(ii)(d): © Qualifications Scotland (SQA). 2022 Higher Physics Paper 2, Q6

Marking: 2022 Paper 2 Q6(c)(ii)(d): © Qualifications Scotland (SQA). 2022 Higher Physics Paper 2, Q6

Exam: Specimen Paper 2 Q4: © Qualifications Scotland (SQA). Specimen Higher Physics Paper 2, Q4

Marking: Specimen Paper 2 Q4: © Qualifications Scotland (SQA). Specimen Higher Physics Paper 2, Q4

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8–9

R Stewart - November 2026